Department of Mechanical Engineering
ME 2201 Kinematics & Dynamics of Machinery
Tutorial 7

Due on: **Oct 5, 2026, In class**

1. For the figure shown, will interference be a problem or not ? Justify.

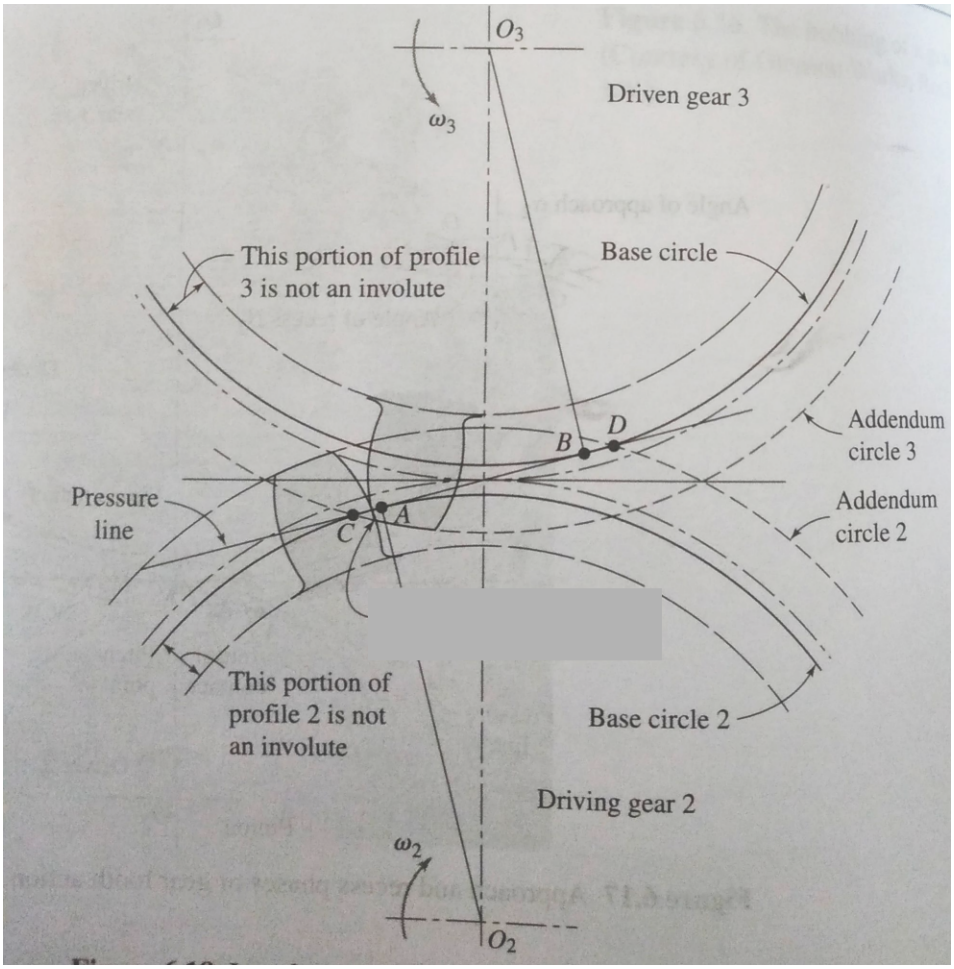


Figure 1: Figure for question 1

2. State if true or false. If true, list disadvantages. Some ways to prevent interference are:

   a. Undercut the tooth.

   b. Use more teeth on the pinion and gear, which increases pitch diameter.

   c. Use a larger pressure angle which creates a smaller base circle.

   d. Make gears with shorther addendum (non-standard).

3. On increasing the center distance (i.e there is now clearnace or backlash between the gears) , what changes ? Answer each sub-question. Refer to figure for clues.

   a Base circle diameters increases ?

   b. Pressure angle increases ?

   c. Pitch circle diameter increases ?

   d. The velocity ratio changes ?

e. Contact ratio remains same/increases/decreases ?

f. Could this (increasing center distance) help resolve problem of interference ?

g. Is fundamental law of gearing satisfied ?

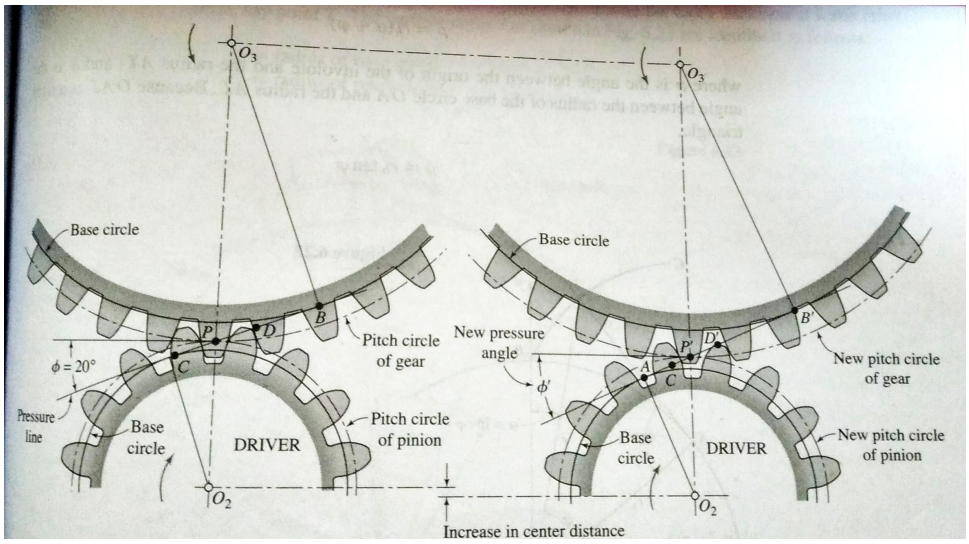


Figure 2: Figure for question 2

4. Two parallel shafts are separated by a distance of 84 mm. They are to be connected by a pair of gears so that output shafet rotates at 40% of input shaft speed. If you need to make a choice of module, pick from table given (note no unique answer!). Check also for undercutting.

Standard Modules SI, mm/tooth	
Preferred	1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 16, 20, 25, 32, 40, 50
Next choice	1.125, 1.375, 1.75, 2.25, 2.75, 3.5, 4.5, 5.5, 7, 9, 11, 14, 18, 22, 28, 36, 45

Figure 3: Figure for question 4

5. Find the module of a pair of gears having 32 and 84 teeth, respectively, whose center distance is 92 mm.

6. The pitch diameters of a pair of mating gears is 90 mm and 210 mm, respectively. If the module is 1.5 mm/tooth, how many teeth are there on each gear.

7. A 6 mm/tooth module, 24-tooth pinion is to drive a 36-tooth gear. The gears are cut on the 20^0 full-depth involute system (standard system). Find and tabulate the addendum, dedendum, clearance (amount by which dedendum of a gear exceeds addendum of mating gear), circular pitch, base pitchm base circle radii, lenth of path of contact. and contact ratio.